



Fractional-Order PID Control for Temperature Regulation of an Alkaline Water Electrolyzer Using Genetic Algorithm-Based Tuning

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Abstract

Alkaline water electrolysis (AWE) is a mature technology for high-purity hydrogen production, but its efficient and safe operation depends critically on accurate temperature regulation. Although higher operating temperatures may improve efficiency, they also accelerate component degradation, whereas lower temperatures reduce hydrogen production performance. Temperature control is therefore a key issue in AWE systems. This paper addresses the temperature regulation problem of an alkaline water electrolyzer through the design of a fractional-order PID (FO-PID) controller. The controller design is based on the transfer function obtained by linearizing a phenomenological nonlinear plant model around a selected operating point. A genetic algorithm-based frequency-domain tuning methodology is proposed, in which the controller parameters are determined so as to satisfy prescribed crossover-frequency and phase-margin specifications, together with additional constraints on the sensitivity and complementary sensitivity functions. The proposed controller is validated in simulation on the original nonlinear model under nominal and disturbed operating conditions, and its performance is compared with that of a classical PID controller and an alternative FO-PID controller tuned using a frequency-domain methodology. The evaluation is carried out in terms of settling time, overshoot, integral of squared error, and disturbance-rejection capability. The results show that the proposed GA-tuned FO-PID controller provides improved closed-loop performance relative to both comparison controllers, indicating that it is a promising alternative for temperature regulation in alkaline water electrolysis systems.

Keywords Alkaline water electrolysis · Temperature control · Fractional-order PID · Genetic algorithm · Frequency-domain tuning

1 Introduction

The growing concern over climate change and greenhouse gas emissions has intensified the search for low-carbon alternatives to fossil-based energy sources such as coal, natural gas, and petroleum (Da Silva Barbosa et al., 2024; de Andrade et al., 2020). In this context, renewable energy technologies, including solar, wind, and hydropower, have received increasing attention due to their potential to reduce carbon emissions and support sustainable development (de Andrade et al., 2014). Nevertheless, the large-scale integration of

renewable sources still faces important challenges, particularly the intermittency of renewable generation and the limited performance of current energy storage technologies, both of which complicate power balancing and reliable grid operation (Gowrisankaran et al., 2016).

Among the available alternatives to mitigate these limitations, green hydrogen has emerged as a promising energy vector due to its high energy density and long-term storage capability (Daoudi & Bounahmidi, 2024). In particular, alkaline water electrolysis (AWE) is an attractive hydrogen production technology because of its technological maturity, robustness, and ability to produce high-purity hydrogen (Daoudi & Bounahmidi, 2024). In addition, AWE can be directly integrated with intermittent renewable sources, enabling the conversion of surplus electrical energy into chemical energy stored as hydrogen (Da Silva Barbosa, 2023; Brauns & Turek, 2020). These characteristics make AWE an important candidate for renewable-based energy systems and power-to-hydrogen applications.

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In AWE systems, temperature is a key operating variable because it directly influences electrochemical conversion efficiency, hydrogen production rate, component degradation, and overall operational safety. The studies in Brauns and Turek (2020); Emam et al. (2024); Amores et al. (2014); Brauns and Turek (2022) indicate that the electrolyzer temperature is commonly operated in the range of 50–80 °C in order to maintain a suitable compromise between efficiency and equipment limitations. In general, low operating temperatures deteriorate the electrochemical performance and reduce hydrogen production efficiency (Qi et al., 2023a), whereas excessively high temperatures may accelerate degradation phenomena and increase material and operating costs (de Groot et al., 2022; Bi et al., 2025). Although higher temperatures may improve thermal efficiency from a purely theoretical standpoint (Jang et al., 2021), practical operation is ultimately constrained by economic considerations, durability requirements, and equipment limitations (Bi et al., 2025). Therefore, maintaining the stack temperature within an appropriate operating range is essential for efficient and reliable electrolyzer operation.

Accurate temperature regulation in AWE systems is challenging due to the nonlinear thermal dynamics of the process, the presence of disturbances, and variations in operating conditions. Under such circumstances, conventional control strategies may yield limited closed-loop performance (Qi et al., 2023a), which has motivated the investigation of more advanced approaches (Liu et al., 2023). For instance, Conker (2019) proposed a fuzzy logic controller to maintain the operating temperature within a suitable range while improving hydrogen production. In David et al. (2021), a robust H_∞ controller based on a linearized model was developed to cope with uncertainties and nonlinear effects. A self-optimizing control strategy based on an efficiency model of the AWE system was proposed in Cheng et al. (2025). Moreover, Qi et al. (2023b) employed a model predictive control strategy that reduced temperature oscillations and enabled operation at higher reference temperatures, with a reported efficiency improvement of approximately 1%. The interested reader is referred to Aguirre et al. (2024) for a detailed review of the main control strategies developed for temperature regulation in alkaline water electrolyzers. Although these approaches have shown promising results, they involve different trade-offs in terms of robustness, model dependence, implementation complexity, and tuning effort, thereby motivating the search for alternative control strategies that combine implementation simplicity with systematic performance-oriented tuning.

Fractional-order representations have been extensively investigated for thermal-system modeling and identification because they can compactly describe the memory and diffusion effects inherent in heat-transfer processes. Foundational studies demonstrated the use of fractional models for

thermal-system identification and thermophysical-parameter estimation (Gabano & Poinot, 2011a, b). Subsequent applications include the identification of building thermal dynamics and biological heat-transfer processes (Chen et al., 2016; Victor et al., 2020). More recent studies have addressed fractional-pole thermal identification in the presence of actuator rate limitations and non-integer formulations for linear and nonlinear inverse heat-conduction problems (Yuan et al., 2022; Battaglia, 2024).

Among the candidate control structures for this problem, fractional-order PID (FO-PID) controllers are particularly attractive because they generalize the integral and derivative actions to non-integer orders, thereby introducing additional degrees of freedom with respect to the classical PID architecture (Jamil et al., 2022). This additional flexibility can be advantageous in systems exhibiting nonlinear behavior, parametric uncertainty, and complex thermal dynamics, for which conventional PID tuning may yield conservative designs or unsatisfactory transient performance (Monje et al., 2010). In this context, several methodologies have been proposed for tuning FO-PID controllers (Padula & Visioli, 2011), including frequency-domain procedures (Li & Gao, 2022), rule-based techniques (Dastjerdi et al., 2018), and optimization-based approaches (Padula & Visioli, 2012; Altbawi et al., 2024). These approaches involve different trade-offs regarding computational effort, achievable performance, robustness specifications, and implementation complexity. Nevertheless, the application of FO-PID control to temperature regulation in alkaline water electrolyzers remains comparatively limited, particularly when controller tuning is formulated to impose explicit frequency-domain performance and robustness requirements (Shah & Agashe, 2016).

The applicability of fractional-order controllers to thermal processes has been demonstrated in a variety of systems. Representative examples include robust FO-PID control of a solar furnace (Beschi et al., 2016), fractional-order temperature control of a fermentation process (Pachauri et al., 2018), and robust temperature regulation of a steel slab reheating furnace with large time-delay uncertainty (Feliu-Batlle & Rivas-Perez, 2019). More recent contributions have considered FO-PID performance assessment in an industrial heating furnace (Li et al., 2020), frequency-domain tuning of FOPI controllers for time-delayed thermal processes (Demiroğlu & Şenol, 2021), and fuzzy fractional-order PID control of a heat exchanger (Al-Dhaifallah, 2023). These studies support the relevance of fractional-order control for thermal systems while also highlighting the need for application-specific tuning and robustness analysis.

Motivated by the above considerations, this paper addresses the temperature regulation problem of an alkaline water electrolyzer using an FO-PID controller tuned through a genetic algorithm-based (GA-based) frequency-domain

methodology. The present study focuses on a constrained formulation in which the controller parameters are selected to satisfy prescribed crossover-frequency and phase-margin specifications, together with constraints on the sensitivity and complementary sensitivity functions. The controller is designed from a transfer function obtained by linearizing a phenomenological nonlinear model around a selected operating point. The cooling system relies on a chiller unit to remove excess heat, while the valve opening is adopted as the manipulated input. The resulting FO-PID controller is evaluated in simulation on the nonlinear plant model and compared with two benchmarks: a classical PID controller tuned using the same GA-based framework and closed-loop specifications, and an FO-PID controller tuned using the frequency-domain method proposed by Monje et al. (2008). The former comparison is intended to isolate the effect of the additional fractional-order degree of freedom, whereas the latter evaluates the proposed GA-based method against an established FO-PID tuning methodology using the same reduced controller structure. A preliminary version of this work was presented in Palomino et al. (2025). The present paper substantially extends that study by introducing the proposed GA-based tuning methodology, which constitutes the main methodological novelty of this contribution.

The remainder of the paper is organized as follows. Section 2 describes the AWE process studied in this work, the control problem, and its mathematical model. Section 3 introduces the classical PID and FO-PID control strategies and the GA-based tuning framework. Section 4 discusses the simulation results and comparative analysis. Finally, Sect. 5 concludes the paper.

2 Description of the Electrolyzer

Figure 1 illustrates the AWE system considered in this work. The stack, composed of multiple electrochemical cells, is the main unit responsible for hydrogen production. The gas–electrolyte mixture leaving the stack is directed to separators, where hydrogen and oxygen are separated from the electrolyte. The remaining electrolyte is then recirculated to the stack, thereby forming a closed-loop process.

Because the electrolysis process is exothermic, the system is equipped with a cooling loop in which cold water is used to remove excess heat. The coolant flow is regulated by a control valve, whose opening is adopted as the manipulated input. The thermal behavior of the electrolyzer is affected by external disturbances, mainly variations in the operating current and, to a lesser extent, changes in ambient temperature (Qi et al., 2023a). Since the heat generation in the stack depends on the electrical operating conditions, current variations directly influence the process temperature. This effect becomes particularly important in applications involv-

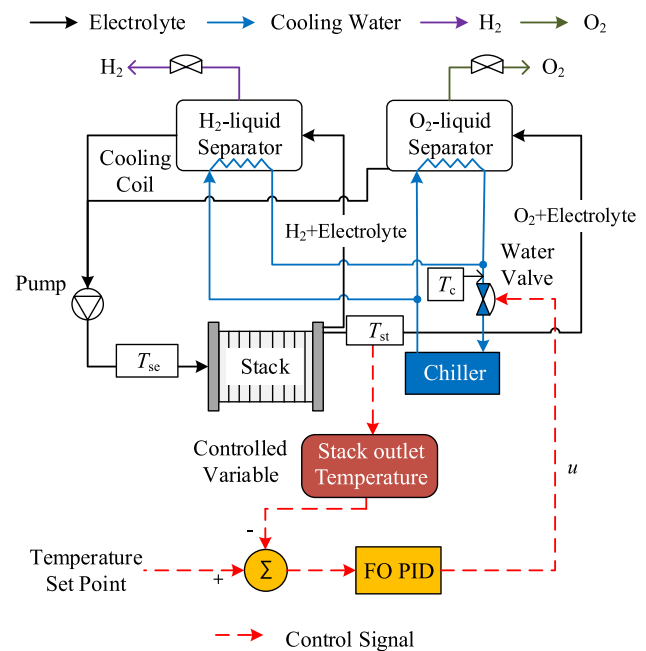


Fig. 1 Schematic diagram of an AWE. Adapted from Qi et al. (2023a)

ing renewable-based power supply, such as photovoltaic or wind generation, but is not restricted to this operating scenario.

2.1 Control Problem

Temperature regulation is essential for ensuring both efficient and safe operation of the electrolyzer. Higher operating temperatures may improve hydrogen production efficiency; however, they also accelerate material degradation and may reduce system reliability. Conversely, low temperatures can impair process performance. For this reason, the electrolyzer is typically operated within the range of 50°C to 90°C (Diéguez et al., 2008; Chi & Yu, 2018).

The control objective is to regulate the stack outlet temperature around a desired reference value in the presence of disturbances, particularly those induced by fluctuations in the electrical current supplied by renewable generation. To this end, temperature sensors installed after the stack provide the feedback signal used by the controller. Based on this measurement, the controller computes the valve-opening command so as to keep the process temperature close to the prescribed setpoint while attenuating the effect of current-induced thermal variations.

2.2 Mathematical Model

To facilitate the analysis and control design, a reduced-order lumped-parameter model is adopted for the AWE system. The model is based on the following assumptions (Qi et

al., 2023a). First, the thermal dynamics are described by three state variables, namely the stack outlet temperature, the separator temperature, and the coolant temperature. Second, spatial temperature distributions within the system components are neglected, so that each subsystem is represented by a spatially uniform temperature. In addition, transport delays are not taken into account.

The coolant flow rate is assumed to be proportional to the valve opening, according to $v_{\text{valve}} = k_v u$, where u denotes the valve aperture (control variable) and k_v is the corresponding proportionality constant. Likewise, the current density is given by $i = I_{\text{cell}}/A_{\text{cell}}$, where I_{cell} is the supplied current and A_{cell} denotes the electrode area.

Under these assumptions, the thermal dynamics of the AWE system are described by the following state equations (Qi et al., 2023a):

$$C_{\text{st}} \frac{dT_{\text{st}}}{dt}(t) = Q_{\text{el}}(t) - Q_{\text{dis,st}}(t) - c_1 v_1 \rho_1 (T_{\text{st}}(t) - T_{\text{se}}(t)), \quad (1)$$

$$C_{\text{se}} \frac{dT_{\text{se}}}{dt}(t) = \frac{1}{2} c_1 v_1 \rho_1 (T_{\text{st}}(t) - T_{\text{se}}(t)) - k_{A_{\text{st}}} \Delta T(t) - Q_{\text{dis,se}}(t), \quad (2)$$

$$C_c \frac{dT_c}{dt}(t) = c_c v_{\text{valve}}(t) \rho_c (T_{\text{c,in}}(t) - T_c(t)) + k_{A_{\text{st}}} \Delta T(t), \quad (3)$$

with

$$Q_{\text{el}}(t) = (U_{\text{cell}}(t) - U_{\text{th}}) I_{\text{cell}}(t) N_{\text{cell}}, \quad (4)$$

$$Q_{\text{dis,st}}(t) = h_e(t) A_{\text{st}} (T_{\text{st}}(t) - T_{\text{amb}}(t)) + \sigma A_{\text{st}} \varepsilon_{\text{st}} (T_{\text{st}}^4(t) - T_{\text{amb}}^4(t)), \quad (5)$$

$$Q_{\text{dis,se}}(t) = \frac{T_{\text{st}}(t) + T_{\text{se}}(t) - 2T_{\text{amb}}(t)}{2R_{\text{se}}}, \quad (6)$$

$$\Delta T(t) = \frac{(T_{\text{st}}(t) - T_c(t))}{\ln\left(\frac{T_{\text{st}}(t) - T_c(t)}{T_{\text{se}}(t) - T_{\text{c,in}}(t)}\right)} - \frac{(T_{\text{se}}(t) - T_{\text{c,in}}(t))}{\ln\left(\frac{T_{\text{st}}(t) - T_c(t)}{T_{\text{se}}(t) - T_{\text{c,in}}(t)}\right)}, \quad (7)$$

$$h_e(t) = 1.3 \left(\frac{T_{\text{st}}(t) - T_{\text{amb}}(t)}{\phi_{\text{st}}} \right)^{0.25}, \quad (8)$$

where U_{cell} , the cell voltage, is modeled as the sum of the reversible voltage and empirical ohmic and activation over-voltage terms, whereas the thermoneutral voltage U_{th} appears only in the heat-generation term Q_{el} .

The cell voltage expression is given by

$$U_{\text{cell}}(t) = s_e \log((t_1 + t_2/\bar{T}(t) + t_3/\bar{T}^2(t))i(t) + 1) + U_{\text{rev}} + (r_1 + r_2 \bar{T}(t))i(t), \quad (9)$$

where s_e , t_1 , t_2 , t_3 , r_1 , and r_2 are empirical constants, A_{st} is the stack surface area, and $\bar{T}$ is defined as

$$\bar{T}(t) = \frac{T_{\text{st}}(t) + T_{\text{se}}(t)}{2}.$$

In Eqs. (1)–(8), T_{st} , T_{se} , and T_c denote the stack outlet temperature, gas–liquid separator temperature, and cooling coil temperature, respectively, and T_{amb} and $T_{\text{c,in}}$ are the ambient and inlet coolant temperature. The term Q_{el} denotes the thermal power generated by the electrochemical reactions in the stack. The heat losses from the stack and separator are represented by $Q_{\text{dis,st}}$ and $Q_{\text{dis,se}}$, respectively. The variable ΔT denotes the temperature difference driving heat transfer among the system components (Qi et al., 2023a). Ultimately, the constant values are C_{st} , C_{se} , C_c , c_1 , v_1 , and ρ_1 , which represent stack heat capacity, separator heat capacity, cooling coil heat capacity, electrolyte thermal capacity, electrolyte flow rate, and electrolyte density, respectively. Similarly, the constants R_{se} , $k_{A_{\text{st}}}$, ρ_c , c_c and ϕ_{st} are the thermal resistance, heat-transfer coefficient, cooling water density, specific heat capacity of the cooling water, and stack diameter, respectively.

The physical parameters adopted in Eqs. (1)–(9) are summarized in Table 1. The considered electrolyzer produces 5 Nm³/h of H₂ while consuming 17.7 kW of electrical power (Qi et al., 2023a).

2.2.1 State-Space Representation

For the state-space formulation, the state vector and the algebraic variable vector are defined as

$$\begin{aligned} x &= [x_1 \ x_2 \ x_3]^T = [T_{\text{st}} \ T_{\text{se}} \ T_c]^T, \\ z &= [z_1 \ z_2 \ z_3 \ z_4]^T = [Q_{\text{el}} \ Q_{\text{dis,st}} \ Q_{\text{dis,se}} \ \Delta T]^T, \\ d &= [d_1 \ d_2 \ d_3]^T = [I_{\text{cell}} \ T_{\text{amb}} \ T_{\text{c,in}}]^T. \end{aligned}$$

With these definitions, the differential part of the model is given by

$$f_1(x, z) = \frac{1}{C_{\text{st}}} \left[z_1(t) - c_1 v_1 \rho_1 (x_1(t) - x_2(t)) - z_2(t) \right],$$

$$f_2(x, z) = \frac{1}{C_{\text{se}}} \left[\frac{1}{2} c_1 v_1 \rho_1 (x_1(t) - x_2(t)) - z_3(t) - k_{A_{\text{st}}} z_4(t) \right],$$

$$f_3(x, z, u, d) = \frac{1}{C_c} \left[c_c k_v u(t) \rho_c (d_3(t) - x_3(t)) + k_{A_{\text{st}}} z_4(t) \right].$$

Table 1 Descriptions and numerical values of the model parameters

Parameter (symbol)	Value
Cell number (N_{cell})	26
Stack diameter (ϕ_{st})	0.61 m
Stack surface area (A_{st})	1.1 m ²
Cell area (A_{cell})	0.196 m ²
Blackness stack surface (ε_{st})	0.8
Heat transfer coefficient (k_{Ast})	140 $\frac{\text{W}}{\text{K}}$
Electrolyte flow rate (v_1)	0.4–0.6 m ³ /hr
Thermal resistance (R_{se})	0.04 $\frac{\text{K}}{\text{W}}$
Stack heat capacity (C_{st})	120 $\frac{\text{kJ}}{\text{K}}$
Separator heat capacity (C_{se})	146 $\frac{\text{kJ}}{\text{K}}$
Cooling coil heat capacity (C_c)	23 $\frac{\text{kJ}}{\text{K}}$
Electrolyte density (ρ_1)	1200 kg/m ³
Electrolyte thermal capacity (c_1)	1000 J/kg K
Stefan–Boltzmann constant (σ)	$5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$
Empirical constant (s_e)	0.16 V
Empirical constant (t_1)	−0.24 m ² /A
Empirical constant (t_2)	26.23 m ² K/A
Empirical constant (t_3)	139.88 $\frac{\text{m}^2 \cdot \text{K}^2}{\text{A}}$
Empirical constant (r_1)	$1.71 \times 10^{-4} \Omega \text{m}^2$
Empirical constant (r_2)	$-1.96 \times 10^{-7} \Omega \text{m}^2/\text{K}$
Reversible voltage (U_{rev})	1.4 V
Thermoneutral voltage (U_{th})	1.55 V
Cooling water density (ρ_c)	997 kg/m ³
Specific heat capacity (c_c)	4.186 kJ/kg K
Valve constant (k_v)	1.1 m ³ /hr

The algebraic part of the model is described by

$$\begin{aligned}
 g_1(x, z, d) &= (U_{\text{cell}}(x, d) - U_{\text{th}})d_1(t)N_{\text{cell}} - z_1(t), \\
 g_2(x, z, d) &= h_e(x, d)A_{\text{st}}(x_1(t) - d_2(t)) - z_2(t) \\
 &\quad + \sigma A_{\text{st}}\varepsilon_{\text{st}}(x_1^4(t) - d_2^4(t)), \\
 g_3(x, z, d) &= \frac{x_1(t) + x_2(t) - 2d_2(t)}{2R_{\text{se}}} - z_3(t), \\
 g_4(x, z, d) &= \frac{(x_1(t) - x_3(t)) - (x_2(t) - d_3(t))}{\ln\left(\frac{x_1(t) - x_3(t)}{x_2(t) - d_3(t)}\right)} \\
 &\quad - z_4(t).
 \end{aligned}$$

Therefore, the nonlinear thermal model can be written in descriptor form as

$$\dot{x}(t) = f(x, z, u, d), \quad (10)$$

$$y(t) = Cx(t) \quad (11)$$

$$0 = g(x, z, d), \quad (12)$$

where

$$\begin{aligned}
 f(x, z, u, d) &= [f_1 \ f_2 \ f_3]^T, \quad C = [1 \ 0 \ 0], \\
 g(x, z, d) &= [g_1 \ g_2 \ g_3 \ g_4]^T.
 \end{aligned}$$

2.2.2 Model Linearization

Let (x^*, z^*, u^*, d^*) be an equilibrium point of the nonlinear descriptor model Eqs. (10)–(12), satisfying $f(x^*, z^*, u^*, d^*) = 0$, and $g(x^*, z^*, u^*, d^*) = 0$. Defining the perturbation variables

$$\begin{aligned}
 \delta x(t) &= x(t) - x^*, & \delta z(t) &= z(t) - z^*, \\
 \delta u(t) &= u(t) - u^*, & \delta d(t) &= d(t) - d^*,
 \end{aligned}$$

the first-order linearization yields

$$\delta \dot{x}(t) = F_x \delta x(t) + F_z \delta z(t) + F_u \delta u(t) + F_d \delta d(t), \quad (13)$$

$$\delta y(t) = C \delta x(t), \quad (14)$$

$$0 = G_x \delta x(t) + G_z \delta z(t) + G_d \delta d(t), \quad (15)$$

where

$$\begin{aligned}
 F_x &:= \left. \frac{\partial f}{\partial x} \right|_{(x^*, z^*, u^*, d^*)}, & F_z &:= \left. \frac{\partial f}{\partial z} \right|_{(x^*, z^*, u^*, d^*)}, \\
 F_u &:= \left. \frac{\partial f}{\partial u} \right|_{(x^*, z^*, u^*, d^*)}, & F_d &:= \left. \frac{\partial f}{\partial d} \right|_{(x^*, z^*, u^*, d^*)}, \\
 G_x &:= \left. \frac{\partial g}{\partial x} \right|_{(x^*, z^*, u^*, d^*)}, & G_z &:= \left. \frac{\partial g}{\partial z} \right|_{(x^*, z^*, u^*, d^*)}, \\
 G_d &:= \left. \frac{\partial g}{\partial d} \right|_{(x^*, z^*, u^*, d^*)}.
 \end{aligned}$$

Assuming that G_z is nonsingular, the algebraic perturbation can be eliminated as

$$\delta z(t) = -G_z^{-1}(G_x \delta x(t) + G_d \delta d(t)).$$

Substituting this expression into Eq. (13) leads to the standard linear state-space model

$$\delta \dot{x}(t) = A \delta x(t) + B \delta u(t) + E \delta d(t), \quad (16)$$

$$\delta y(t) = C \delta x(t), \quad (17)$$

with

$$A = F_x - F_z G_z^{-1} G_x, \quad B = F_u,$$

$$E = F_d - F_z G_z^{-1} G_d.$$

Substituting the parameter values given in Table 1 into Eqs. (10)–(12), and selecting the operating point

$$x^* = [82.24 \ 68.11 \ 49.29]^T,$$

$$d^* = [500 \ 25 \ 40]^T,$$

$$z^* = [9505.6 \ 202.78 \ 1046.18 \ 25.75]^T,$$

$$u^* = 0.5,$$

which corresponds to a thermal equilibrium of the nonlinear system Eqs. (10)–(12), the linear state-space model Eqs. (16), (17) is obtained with

$$A = \begin{bmatrix} -7.05 & 6.61 & 0 \\ 2.25 & -3.39 & 0.45 \\ 2.88 & 3.21 & -22.84 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ -1854.49 \end{bmatrix},$$

$$E = \begin{bmatrix} 0.2171 & 0.044 & 0 \\ 0 & 0.17 & 0.54 \\ 0 & 0 & 96.33 \end{bmatrix}.$$

Moreover, the transfer function from the control input δu to the output δy is given by

$$G(s) = \frac{-5583}{s^3 + 33.3s^2 + 246.3s + 186.5}. \quad (18)$$

For the numerical simulations, the nonlinear plant model Eqs. (10)–(12) is evaluated in discrete time using the Euler integration method.

3 Fractional-Order PID Controller

Figure 2 depicts the control configuration, where r_d , u , y , and d denote the temperature reference, the control input (valve opening), the measured output (T_{st}), and the disturbance, respectively. Based on this configuration, the present section addresses the design of the proposed FO-PID controller. The controller parameters are determined through a genetic algorithm-based optimization procedure formulated in the frequency domain. More specifically, the tuning is carried out to satisfy prescribed crossover-frequency and phase-margin specifications, together with additional constraints on the sensitivity and complementary sensitivity functions. This yields a systematic design framework in which performance and robustness requirements are explicitly incorporated into the controller synthesis.

The FO-PID controller generalizes the classical PID structure by allowing non-integer orders in the integral and derivative actions. Its continuous-time transfer function is given by

$$C(s) = K_p + \frac{K_i}{s^\alpha} + K_d s^\alpha, \quad (19)$$

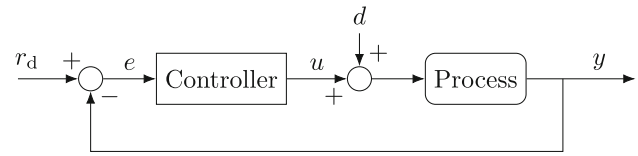


Fig. 2 Closed-loop control configuration

where $\alpha \in (0, 1)$ denotes the fractional order. Compared with the classical PID controller, the FO-PID structure introduces an additional tuning degree of freedom through α , which can be exploited to improve the shaping of the closed-loop response. A more general FO-PID structure may employ different fractional orders for the integral and derivative actions (Shah & Agashe, 2016). In the present study, a reduced structure with a common fractional order α is adopted to limit the number of optimization variables and simplify the tuning and discrete-time implementation. This choice also allows the analysis to focus on the performance improvement obtained by introducing a single additional fractional-order parameter relative to the classical PID controller.

For digital implementation, the fractional operator s^α can be realized using different numerical or data-driven approaches, including rational transfer-function approximations (Li & Cai, 2019) and neural network-based representations of fractional-order dynamics (Puchalski & Rutkowski, 2020). In this work, Muir's recursive approximation is adopted (Vinagre et al., 2003), since it yields a rational representation with moderate computational complexity and provides increasingly accurate approximations as the recursion order increases over the frequency range of interest. In general, the approximation is written as

$$s^\alpha \approx \left(\frac{2}{T_s} \right)^\alpha \frac{A_n(z^{-1}, \alpha)}{A_n(z^{-1}, -\alpha)}, \quad (20)$$

where $A_n(\cdot, \alpha)$ is a polynomial obtained recursively.

In this study, a third-order approximation ($n = 3$) is adopted, leading to

$$s^\alpha \approx \left(\frac{2}{T_s} \right)^\alpha \frac{z^3 - \alpha z^2 + \frac{\alpha^2}{3} z - \frac{\alpha}{3}}{z^3 + \alpha z^2 + \frac{\alpha^2}{3} z + \frac{\alpha}{3}}. \quad (21)$$

For notational convenience, define

$$c = \left(\frac{2}{T_s} \right)^\alpha,$$

$$p(z) = z^3 - \alpha z^2 + \frac{\alpha^2}{3} z - \frac{\alpha}{3},$$

$$q(z) = z^3 + \alpha z^2 + \frac{\alpha^2}{3} z + \frac{\alpha}{3}.$$

Substituting Eq. (21) into Eq. (19) yields the discrete-time transfer function

$$C(z) = \frac{K_p p(z) q(z) + K_i c^{-1} q^2(z) + K_d c p^2(z)}{p(z) q(z)}. \quad (22)$$

Since both $p(z)$ and $q(z)$ are third-order polynomials, the resulting controller is represented by a sixth-order rational discrete-time transfer function.

It should be noted that the finite-order Muir approximation does not preserve an exact pole at $z = 1$ for $0 < \alpha < 1$, as can be seen in Eq. (22). Consequently, the approximated fractional integral action has finite gain at zero frequency, and exact zero steady-state error for constant references is not structurally guaranteed. Nevertheless, the maximum residual steady-state error observed in the numerical simulations was around 0.0035°C , which is negligible for the considered operating conditions.

Importantly, the finite-order Muir approximation Eq. (20) replaces the infinite-dimensional fractional operator s^α with a rational discrete-time transfer function. Consequently, the characteristic power-law magnitude response and constant-phase behavior of the fractional operator are reproduced only approximately (Puchalski & Rutkowski, 2020). In general, increasing the approximation order improves the representation of the fractional operator over the frequency range of interest, but also increases the order and computational complexity of the resulting discrete-time controller. In the present work, a third-order approximation is adopted as a compromise between approximation fidelity and implementation complexity. Therefore, the obtained controller should be interpreted as a finite-dimensional approximation of the ideal FO-PID controller.

3.1 Genetic Algorithm-Based Tuning

The tuning of PID controllers, including both classical and fractional-order structures, is often based on empirical rules, analytical formulas, or trial-and-error procedures (Aström & Hägglund, 1995). While some classical methods can achieve satisfactory performance and may target particular robustness or transient response requirements, they are usually tailored to specific assumptions and are less convenient when several design specifications must be imposed simultaneously. This issue is more pronounced for FO-PID controllers, whose additional degrees of freedom increase the complexity of the tuning problem.

To address this issue, this work adopts a GA-based tuning procedure. GAs are population-based optimization methods particularly well suited to nonlinear, nonconvex, and multi-parameter problems, and are less sensitive to local minima than gradient-based local search methods (Jamil et al., 2022). In the present context, the controller parameters

are treated as decision variables, and the tuning problem is formulated as a constrained optimization problem in which the objective function and constraints are selected to enforce the desired closed-loop behavior. In particular, the optimization incorporates frequency-domain specifications associated with stability margins, robustness, and closed-loop performance. As with any metaheuristic procedure, the quality of the obtained solution depends on the definition of the admissible parameter set and on the choice of the algorithm hyperparameters, such as population size, crossover probability, and mutation probability (Eltayeb Taha et al., 2024).

The methodology adopted in this paper is based on a frequency-domain formulation in which the controller parameters are optimized so as to satisfy prescribed crossover and phase-margin requirements, together with additional constraints on the sensitivity and complementary sensitivity functions. In this way, the search carried out by the GA is guided by explicit robustness and performance specifications, leading to a systematic tuning procedure for both PID and FO-PID controllers. This approach is inspired by the frequency-domain tuning strategy presented in Monje et al. (2008).

3.1.1 Cost Function and Frequency-Domain Constraints

Consider the unity-feedback configuration shown in Fig. 2 with plant transfer function $G(s)$ and controller $C(s, \theta)$, where

$$\theta := [K_p \ K_i \ K_d \ \alpha]^\top,$$

is the corresponding decision-variable vector. Define the open-loop transfer function as

$$L(s, \theta) := C(s, \theta) G(s). \quad (23)$$

The associated sensitivity and complementary sensitivity functions are defined, respectively, as

$$S(s, \theta) := \frac{1}{1 + L(s, \theta)},$$

$$T(s, \theta) := \frac{L(s, \theta)}{1 + L(s, \theta)}.$$

The tuning objective is to impose a prescribed gain crossover frequency $\omega_{cg} > 0$. Since the gain crossover frequency is characterized by $|L(j\omega_{cg}, \theta)| = 1$, the cost function is chosen as the deviation of the open-loop magnitude from 0 dB at ω_{cg} , namely

$$J(\theta) := |20 \log_{10} |L(j\omega_{cg}, \theta)||. \quad (24)$$

Importantly, the absolute value in Eq. (24) ensures that the optimization penalizes any deviation from the desired 0 dB crossover condition. Otherwise, the minimization could incorrectly favor very negative values instead of enforcing crossover.

Equivalently, using the notation $|H(j\omega)|_{dB} := 20 \log_{10} |H(j\omega)|$, one may rewrite Eq. (24) as

$$J(\theta) = \left| |C(j\omega_{cg}, \theta)G(j\omega_{cg})|_{dB} \right|. \quad (25)$$

To enforce a prescribed phase margin $\varphi_m^* \in (0, \pi)$, the phase of the open-loop transfer function at ω_{cg} is constrained by

$$\angle L(j\omega_{cg}, \theta) = -\pi + \varphi_m^*, \quad (26)$$

where $\angle(\cdot)$ denotes the continuous phase angle. When the crossover condition $J(\theta) = 0$ is satisfied, Eq. (26) is equivalent to imposing the phase margin φ_m^* at the prescribed crossover frequency.

Additional closed-loop performance and robustness requirements are imposed through frequency-domain bounds on T_k and S_k . First, the complementary sensitivity function is constrained as

$$|T(j\omega, \theta)|_{dB} \leq C_1, \quad \forall \omega \in \Omega_T, \quad (27)$$

where $\Omega_T \subset \mathbb{R}_{\geq 0}$ is a prescribed frequency range and C_1 is a design parameter. This constraint limits the amplification of the closed-loop transfer function, and is commonly used to shape bandwidth and attenuate measurement-noise transmission at high frequencies.

Second, the sensitivity function is required to satisfy

$$|S(j\omega, \theta)|_{dB} \leq C_2, \quad \forall \omega \in \Omega_S, \quad (28)$$

where $\Omega_S \subset \mathbb{R}_{\geq 0}$ is another prescribed frequency range and C_2 is a design parameter. This bound is used to limit the effect of output disturbances and to improve robustness with respect to plant uncertainty. Since

$$S(s, \theta) + T(s, \theta) = 1,$$

the constraints Eqs. (27), (28) capture the classical trade-off between performance and robustness.

The tuning problem is then formulated as

$$\min_{\theta \in \Theta} J(\theta) \quad (29)$$

s.t.

$$\varphi_{min} \leq \pi + \angle L(j\omega_{cg}, \theta) \leq \varphi_{max}, \quad (30)$$

$$|T(j\omega, \theta)|_{dB} \leq C_1, \quad \forall \omega \in \Omega_T, \quad (31)$$

$$|S(j\omega, \theta)|_{dB} \leq C_2, \quad \forall \omega \in \Omega_S, \quad (32)$$

where Θ denotes the admissible set of controller parameters, defined by the chosen lower and upper bounds for each decision variable.

For a fair comparison, the classical PID and FO-PID controllers are tuned using the same GA-based optimization framework. Both designs employ the same prescribed crossover-frequency and phase-margin requirements, the same sensitivity and complementary sensitivity constraints, and the same GA hyperparameters. The only difference is the controller parameterization: the classical PID optimization uses K_p , K_i , and K_d as decision variables, whereas the FO-PID optimization additionally includes the fractional order α .

Table 2 summarizes the hyperparameters adopted in the optimization procedure. No user-defined initial population is specified; therefore, the initial population is generated randomly by the algorithm. The constraint tolerance is set to 10^{-5} . For both the classical PID and FO-PID controllers, the gains K_p , K_i , and K_d are constrained to the interval $[-10, 10]$, while the fractional order of the FO-PID controller is restricted to $0.001 \leq \alpha \leq 0.999$. The GA hyperparameters were selected through preliminary numerical tests to provide a suitable compromise between convergence reliability and computational effort. Using these settings, MATLAB's ga solver determines the controller parameters according to the proposed constrained optimization formulation. The same GA hyperparameters and gain bounds are adopted for both controllers to ensure a fair comparison.

4 Results

This section evaluates the proposed FO-PID controller and compares its performance with a classical PID controller. Five simulation scenarios are considered: nominal setpoint tracking, disturbance rejection, varying operating conditions, operation under a measured solar-current profile, and operation under measurement noise. In the first two cases, the nominal operating conditions are fixed at $I_{cell} = 500$ A, $T_{amb} = 25^\circ\text{C}$, and $T_{c,in} = 40^\circ\text{C}$.

All closed-loop simulations are performed on the nonlinear plant model Eqs. (10)–(12) in MATLAB, with sampling period $T_s = 0.01$ h. The GA-based frequency-domain tuning is carried out using the continuous-time linearized transfer function $G(s)$ and the continuous-time controller structure $C(s)$. After the controller parameters are obtained, all controllers are implemented in discrete time. For the FO-PID controllers, the fractional operators are realized through Muir's rational approximation given in Eq. (22).

The prescribed crossover frequency is $\omega_{cg} = 12$, and the phase margin is constrained to satisfy $30^\circ \leq \varphi_m \leq 70^\circ$. The sensitivity and complementary sensitivity constraints are imposed over $\Omega_S = [0, 1.2]$ and $\Omega_T = [120, \infty)$, respec-

Table 2 Genetic algorithm hyperparameters used in the controller tuning procedure

Parameter	FO-PID _{GA}	PID
Decision variables	$[K_p, K_i, K_d, \alpha]$	$[K_p, K_i, K_d]$
Number of variables	4	3
Population size	50	50
Max. generations	400	400
Initial population	Random	Random
Constraint tolerance	10^{-5}	10^{-5}
Crossover fraction	80%	80%
Max. stall generations	50	50
Lower bounds	$[-10, -10, -10, 10^{-3}]$	$[-10, -10, -10]$
Upper bounds	$[10, 10, 10, 0.999]$	$[10, 10, 10]$

tively, with bounds $C_1 = C_2 = -20$ dB. The same GA-based tuning procedure and frequency-domain specifications are used for both controllers. For the classical PID benchmark, the admissible set Θ is restricted to K_p , K_i , and K_d .

The resulting PID parameters are $K_p = -0.125$, $K_i = -0.869$, $K_d = -0.046$. The corresponding continuous-time closed-loop poles are $s_{1,2} = -16.05 \pm 14.01i$, and $s_{3,4} = -0.60 \pm 3.21i$. Since all poles have negative real parts, the linearized closed-loop system with the classical PID controller is asymptotically stable.

For the FO-PID_{GA} controller, the tuned parameters are $K_p = -0.58$, $K_i = -7.56$, $K_d = -0.241$, and $\alpha = 0.88$. The corresponding closed-loop poles are $s_{1,2} = -11.62 \pm 27.57i$, and $s_{3,4} = -2.42 \pm 6.43i$. All these roots have negative real parts, confirming the stability of the continuous-time FO-PID_{GA} closed loop.

Since the FO-PID_{GA} controller is implemented using Muir's rational approximation, the stability of the resulting discrete-time linearized closed-loop system was also verified. Its closed-loop poles are listed below

$$\begin{aligned} z_1 &= -0.93, \\ z_{2,3} &= 0.91 \pm 0.11i, \\ z_{4,5} &= 0.88 \pm 0.21i, \\ z_{6,7} &= -0.029 \pm 0.55i, \\ z_{8,9} &= 0.029 \pm 0.55i. \end{aligned}$$

Since the modulus of every pole is smaller than one, the discrete-time linearized closed-loop system is stable.

For an additional comparison, the proposed FO-PID controller is evaluated against the frequency-domain tuning methodology proposed by Monje et al. (2008). The original method considers five independent controller parameters. In the present work, it is adapted by imposing the same fractional order for the integral and derivative actions, resulting in the same four-parameter structure used by the proposed controller. The deviation of the open-loop magnitude from (0) dB

at the prescribed crossover frequency is minimized, while the phase-margin requirement, the zero phase-slope condition at the crossover frequency, and the bounds on the sensitivity and complementary sensitivity functions are imposed as nonlinear constraints. The optimization is performed using MATLAB's *fmincon* solver with the sequential quadratic programming algorithm. The controller gains are restricted to the interval $[-10, 10]$ and $\alpha \in [10^{-3}, 0.999]$. The design specifications are $\omega_{cg} = 12$ rad/h, $\varphi_m = 60^\circ$, $|S(j\omega)|_{dB} \leq -20$ dB for $\omega \in [0, 1.2]$, and $|T(j\omega)|_{dB} \leq -20$ dB for $\omega \in [120, \infty)$. To reduce sensitivity to the initial point, 50 randomly generated feasible starting points are considered, with a maximum of 400 iterations and a constraint tolerance of 10^{-5} . The feasible solution with the lowest objective-function value is retained and denoted by FO-PID_{MV}. This controller is discretized using the same third-order Muir approximation and sampling period employed for the proposed FO-PID_{GA}. The resulting gains are $K_p = -0.26$, $K_i = -0.59$, $K_d = -0.004$, and $\alpha = 0.94$. In this case, the closed-loop poles are $s_{1,2} = -2.4246 \pm 6.37i$.

For FO-PID_{MV}, the corresponding poles are

$$\begin{aligned} z_1 &= -0.97, \\ z_2 &= 0.93, \\ z_3 &= 0.77, \\ z_{4,5} &= -0.015 \pm 0.57i \\ z_{6,7} &= 0.015 \pm 0.57i, \\ z_{8,9} &= 0.9778 \pm 0.0696i. \end{aligned}$$

4.1 Setpoint Tracking Performance

This subsection evaluates the setpoint-tracking performance of the classical PID and FO-PID controllers under nominal operating conditions. The comparison focuses on tracking accuracy and transient response characteristics.

Figure 3 presents the closed-loop temperature responses for successive reference values of 65°C , 70°C , and 75°C .

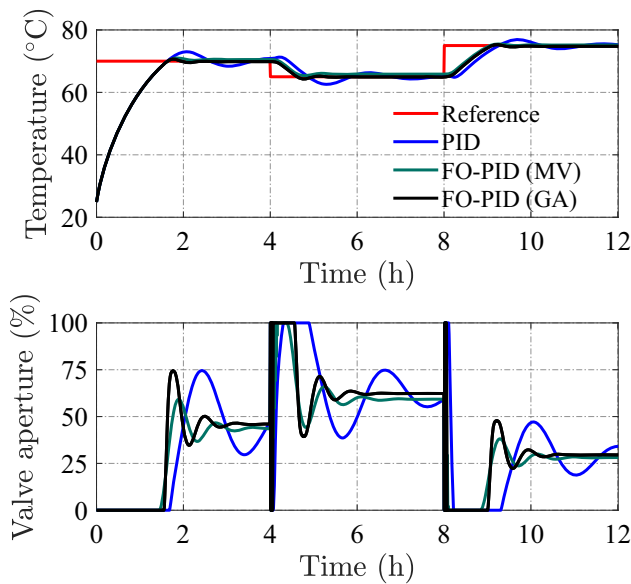


Fig. 3 Closed-loop setpoint-tracking responses of the classical PID and FO-PID controllers

The total simulation time is 12 h, and each reference value is maintained for 4 h. PID and both FO-PID controllers track the prescribed reference values. However, the classical PID controller produces more pronounced overshoots and slower transient responses, whereas the FO-PID_{GA} controller provides better-damped responses and shorter settling times. The FO-PID_{MV} controller also improves the transient response relative to the classical PID.

Table 3 summarizes the performance indices, namely the settling time, the integral of squared error¹ (ISE), and the percent overshoot obtained for each setpoint. The FO-PID_{GA} controller yields lower ISE values, shorter settling times, and smaller overshoots for all three reference values when compared with the classical PID controller. Its average settling time is 1.56 h, compared with 3.41 h for the classical PID controller and 1.70 h for the FO-PID_{MV}, corresponding to reductions of approximately 54.3% and 8.23%, respectively. Furthermore, the largest overshoot produced by the classical PID controller is 4.25%, whereas the maximum overshoot obtained with the FO-PID_{GA} controller is 1.23%, compared with 1.42% for the FO-PID_{MV}. However, at the 75°C setpoint, the ISE of the FO-PID_{GA} controller is 39.94 °C²h, which is higher than the value of 32.70 °C²h obtained with the FO-PID_{MV} controller. These results indicate that the FO-PID_{GA} controller provides faster and better-damped tracking responses over the considered setpoint range.

In terms of control effort, two indices are considered to assess the actuator demand. Table 4 summarizes these indices, where $\Delta u(k) = u(k) - u(k-1)$, $\sum_{k=1}^N |\Delta u(k)|$

represents the accumulated valve-command variation, and $\max_k |\Delta u(k)|$ represents the largest variation between two consecutive commands.

As shown, both fractional-order controllers require a larger accumulated valve-command variation than the classical PID controller. Specifically, the accumulated variation increases from 861.14 for the PID controller to 1848.52 and 1660.64 for the FO-PID_{MV} and FO-PID_{GA}, respectively. The maximum command increment is equal to 100 for all three controllers; therefore, the FO-PID_{GA} does not produce a larger peak increment than the classical PID in this test.

4.2 Disturbance Rejection Performance

This subsection evaluates the disturbance-rejection capability of the classical PID and FO-PID controllers. In the considered application, variations in the electrolyzer current constitute the main disturbance source, since the electrical power supplied to the system may originate from intermittent renewable sources such as solar panels and wind turbines. By contrast, the ambient temperature and the coolant inlet temperature are expected to exhibit only small variations around their nominal values. For this reason, the temperature reference is fixed at 70°C, and a step disturbance is introduced at $t = 0.5$ h through a 50 A decrease in the electrolyzer current with respect to its nominal operating value.

Figure 4 shows the corresponding closed-loop responses. All three controllers are able to reject the disturbance and drive the temperature back to the reference. However, the FO-PID_{GA} controller yields a faster recovery and a less pronounced transient deviation.

Table 5 summarizes the resulting performance indices. The FO-PID_{GA} controller produces the smallest undershoot, with a value of 1.08°C, compared with 1.14°C for the classical PID and 1.19°C for the FO-PID_{MV}. It also provides the shortest recovery time, reducing it from 1.82 h and 1.33 h for the PID and FO-PID_{MV}, respectively, to 0.75 h. These values correspond to reductions of 58.79% and 43.61%, respectively. Furthermore, the FO-PID_{GA} provides the lowest post-disturbance ISE, with a value of 0.25 °C²h, compared with 0.68 °C²h for the PID and 0.45 °C²h for the FO-PID_{MV}. These results demonstrate its improved disturbance-rejection performance under the considered operating conditions.

As shown in Table 6, the FO-PID_{MV} controller presents the largest accumulated valve-command variation, with 159.95%, compared with 112.89% and 122.96% for the PID and FO-PID_{GA} controllers, respectively. This indicates that FO-PID_{MV} requires the greatest overall actuator activity during the disturbance-rejection test. However, FO-PID_{MV} presents the smallest maximum command increment, with 13.28%. In contrast, the PID and FO-PID_{GA} controllers exhibit similar maximum increments of 45.81% and 45.01%, respectively. Therefore, FO-PID_{GA} requires slightly greater

¹ The ISE is defined as $\int_{t_a}^{t_b} e^2(\tau) d\tau$, where e denotes the error between the reference signal and the system output, and $t_a < t_b$.

Table 3 Setpoint-tracking performance indices under nominal operating conditions

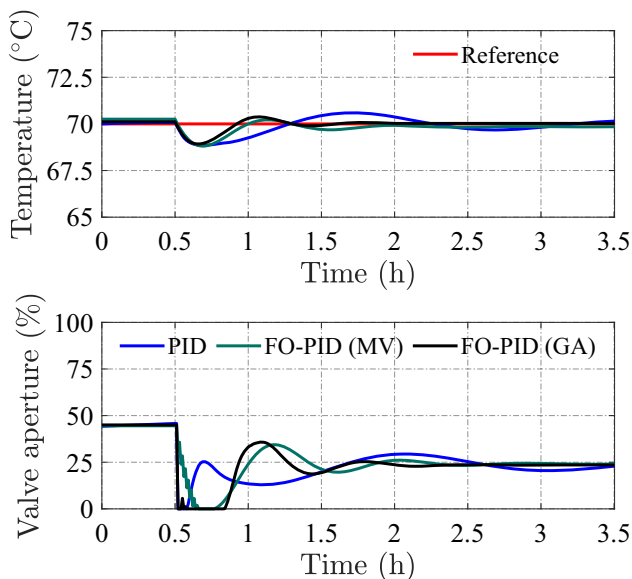
Controller	Setpoint (°C)	Overshoot (%)	Settling time (h)	ISE (°C ² h)
PID	65	3.69	3.20	5.73
	70	4.25	3.58	22.85
	75	0.60	3.45	110.22
FO-PID _{MV}	65	1.42	2.00	1.92
	70	0.00	1.58	16.68
	75	0.67	1.51	32.70
FO-PID _{GA}	65	1.23	1.17	0.27
	70	0.85	2.21	8.87
	75	0.36	1.31	39.94

Table 4 Control-effort variation indices for the valve-opening command during setpoint tracking

Controller	$\sum_{k=1}^N \Delta u(k) $ (%)	$\max(\Delta u(k))$ (%)
PID	861.14	100
FO-PID _{MV}	1848.52	100
FO-PID _{GA}	1660.64	100

Table 6 Control-effort variation indices for the valve-opening command during the disturbance-rejection test

Controller	$\sum_{k=1}^N \Delta u(k) $ (%)	$\max(\Delta u(k))$ (%)
PID	112.89	45.81
FO-PID _{MV}	159.95	13.28
FO-PID _{GA}	122.96	45.01

**Fig. 4** Closed-loop responses of the classical PID and FO-PID controllers to a 50 A step decrease in the electrolyzer current**Table 5** Performance indices for the step disturbance in the electrolyzer current

Controller	Undershoot (°C)	Recovery time (h)	ISE (°C ² h)
PID	1.14	1.82	0.68
FO-PID _{MV}	1.19	1.33	0.45
FO-PID _{GA}	1.08	0.75	0.25

accumulated valve movement than the classical PID but produces a comparable peak command increment. Such peak variations may require rapid valve movement and should therefore be checked against the stroke-rate limitation of the physical actuator.

4.3 Robustness Assessment under Varying Operating Conditions

The frequency-domain specifications employed in the controller design, including the prescribed phase margin and the constraints on the sensitivity and complementary sensitivity functions, provide robustness requirements around the nominal linearization point. To assess whether the resulting controllers preserve satisfactory performance away from this nominal condition, additional simulations were conducted on the nonlinear plant model under variations in the electrolyzer current and coolant inlet temperature. The controller parameters were kept fixed throughout all tests; hence, no retuning was performed when the operating conditions changed.

The stack-temperature reference and ambient temperature were maintained at $T_{\text{ref}} = 70^\circ\text{C}$, and $T_{\text{amb}} = 25^\circ\text{C}$, whereas the electrolyzer current and coolant inlet temperature were varied within the sets $I_{\text{cell}} \in \{500, 550, 650\}$ A, and $T_{\text{c,in}} \in \{35, 40, 45\}^\circ\text{C}$, respectively. The sequence of operating conditions is summarized in Table 7.

Figure 5 presents the corresponding closed-loop responses. All controllers maintain stable operation and regulate the stack temperature around the prescribed reference for all

Table 7 Operating conditions used in the robustness assessment

Time interval (h)	I_{cell} (A)	$T_{\text{c,in}}$ (°C)
$0 \leq t < 4$	500	35
$4 \leq t < 6$	650	35
$6 \leq t < 8$	550	35
$8 \leq t < 10$	500	40
$10 \leq t < 12$	650	40
$12 \leq t < 14$	550	40
$14 \leq t < 16$	500	45
$16 \leq t < 18$	650	45
$18 \leq t \leq 20$	550	45

tested combinations of I_{cell} and $T_{\text{c,in}}$. The FO-PID_{GA} controller restores the output to the reference more rapidly after most operating-point variations, although the relative magnitude of the transient temperature deviation depends on the particular transition.

The quantitative comparison is based on the maximum absolute temperature deviation, recovery time, and ISE associated with each operating-point transition. The maximum absolute deviation is measured relative to the reference temperature, while the recovery time is defined as the time required for the output to return to and remain within a $\pm 1.65\%$ band around the reference. The ISE is calculated

separately over each operating-condition interval. Table 8 summarizes the resulting performance indices.

The FO-PID_{GA} controller provides a shorter recovery time than the classical PID controller in seven of the eight operating-point transitions and achieves the shortest recovery time among all three controllers in five transitions. Averaged over all transitions, the recovery time decreases from 1.43 h and 1.25 h for the classical PID and FO-PID_{MV} controllers, respectively, to 0.97 h for the FO-PID_{GA} controller. Additionally, the FO-PID_{GA} controller achieves the lowest average ISE, with a value of $0.76^\circ\text{C}^2\text{h}$, compared with $0.85^\circ\text{C}^2\text{h}$ for the PID controller and $2.91^\circ\text{C}^2\text{h}$ for the FO-PID_{MV} controller.

However, the maximum temperature deviations vary according to the operating-point transition, and their average value increases from 1.16°C for the PID controller to 1.52°C for the FO-PID_{GA} controller, while remaining below the 1.84°C obtained with the FO-PID_{MV} controller. Therefore, the main advantages of the FO-PID_{GA} controller in this assessment are its faster recovery and lower average ISE, rather than a uniformly smaller transient deviation.

These results provide numerical evidence that the fixed FO-PID_{GA} controller preserves satisfactory regulation performance over the considered operating region, despite being designed from a linearized model obtained at a single nominal operating point. Nevertheless, this evaluation should be

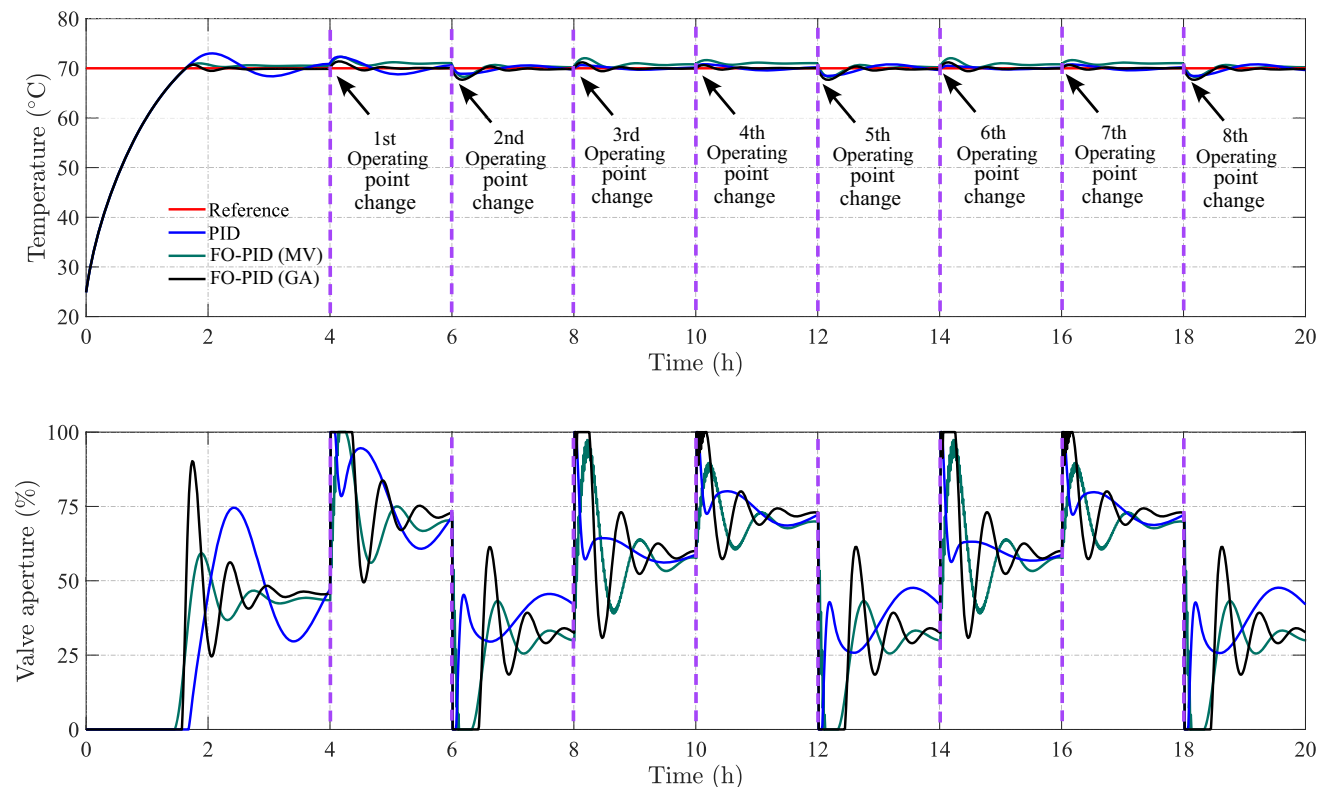
**Fig. 5** Closed-loop responses at different operating points for the same temperature reference of 70°C

Table 8 Performance indices under varying operating conditions

Trans. time (h)	PID				FO-PID _{MV}				FO-PID _{GA}			
	Max. dev. (°C)	Rec. time (h)	ISE (°C ² h)	Max. dev. (°C)	Rec. time (h)	ISE (°C ² h)	Max. dev. (°C)	Rec. time (h)	Max. dev. (°C)	Rec. time (h)	ISE (°C ² h)	Max. dev. (°C)
4	2.33	1.77	2.72	2.30	1.35	4.87	1.39	0.62	1.39	0.62	0.95	1.39
6	1.12	1.44	0.69	1.72	1.09	1.12	2.31	1.03	2.31	1.03	1.00	2.31
8	0.63	0.76	0.16	2.05	1.37	3.38	1.15	0.70	1.15	0.70	0.62	1.15
10	0.82	0.88	0.32	1.66	1.39	4.16	0.81	0.62	0.81	0.62	0.42	0.81
12	1.54	1.60	1.23	1.73	0.99	1.12	2.32	1.18	2.32	1.18	1.02	2.32
14	0.51	1.69	0.11	2.06	1.36	3.37	1.20	1.81	1.20	1.81	0.63	1.20
16	0.79	1.52	0.30	1.66	1.38	4.16	0.63	0.65	0.63	0.65	0.42	0.63
18	1.54	1.81	1.24	1.55	1.05	1.11	2.33	1.18	2.33	1.18	1.02	2.33

Table 9 Control-effort variation indices for the valve actuator under varying operating conditions

Controller	$\sum_{k=1}^N \Delta u(k) $ (%)	$\max(\Delta u(k))$ (%)
PID	1162.38	72.21
FO-PID _{MV}	2015.18	24.93
FO-PID _{GA}	1893.54	72.99

interpreted as a simulation-based robustness assessment over the tested operating range, rather than as a formal proof of global robustness for the nonlinear plant.

The control-effort variation indices for the varying operating-condition test are summarized in Table 9. The accumulated valve-command variation is 1162%, 2015%, and 1893% for the PID, FO-PID_{MV}, and FO-PID_{GA} controllers, respectively. Thus, FO-PID_{GA} requires greater accumulated valve movement than the classical PID, which may contribute to increased actuator wear over long operating periods, but requires less accumulated movement than FO-PID_{MV}. The maximum command increments are 72.21%, 24.93%, and 72.99%, respectively. From a practical point of view, these command changes should be checked against the stroke-rate limitation of the physical valve.

4.4 Evaluation under a Measured Solar-Current Profile

An additional evaluation is conducted using a measured current profile, depicted in Fig. 6, obtained from a photovoltaic source. The profile presents the typical diurnal pattern of solar generation, with a maximum current of 726 A occurring at approximately $t = 6$ h. Short-term fluctuations are observed between $t = 2$ h and $t = 5$ h, which are attributed to transient variations in solar irradiance caused by passing clouds.

The measured current profile is then applied to the nonlinear plant model while all controllers are required to regulate the electrolyzer temperature around the reference 70°C. The corresponding closed-loop responses are shown in Fig. 7. During the initial and final portions of the experiment, all controllers exhibit similar behavior, since the available solar current is low and the corresponding control action remains inactive or negligible. The main differences arise during the interval in which the solar current reaches higher values and the cooling action becomes effective.

The quantitative comparison is based on the percent overshoot, settling time, and ISE. The ISE is computed over the interval from 4 h to 10 h, which corresponds to the period in which the solar current is sufficiently high to produce a significant thermal response.

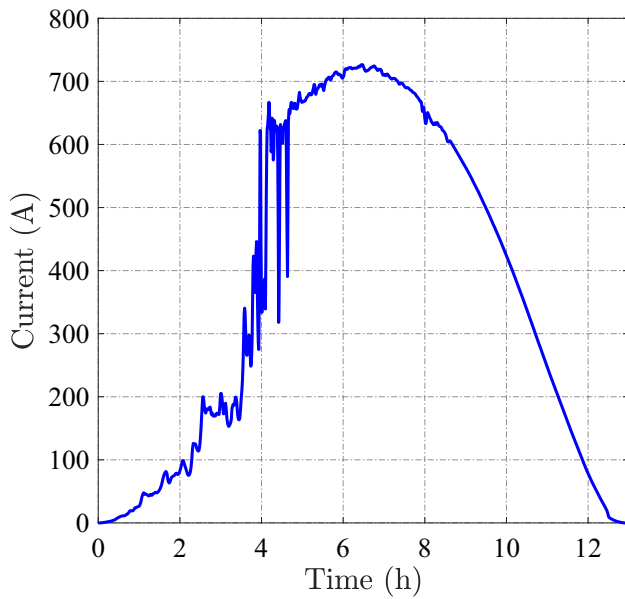


Fig. 6 Measured solar-current profile used in the simulations

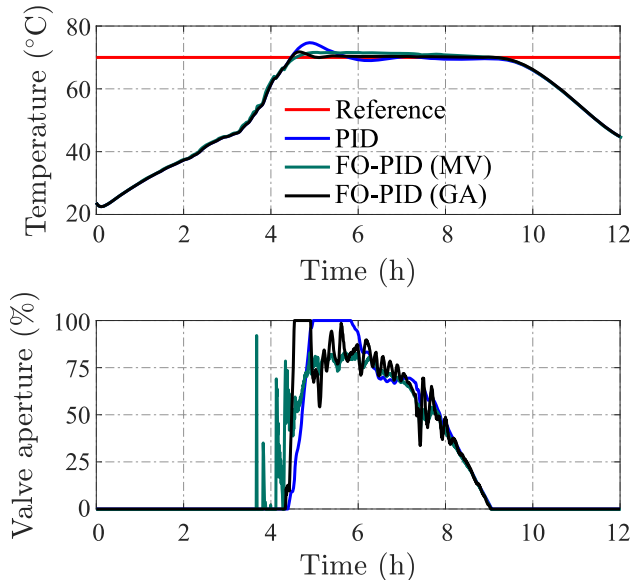


Fig. 7 Closed-loop responses of the PID and FO-PID controllers under the measured solar-current profile

The classical PID controller reaches a maximum temperature of 73.75°C, corresponding to an overshoot of 3.75°C, or 5.34%, relative to the 70°C reference. In comparison, the FO-PID_{GA} controller produces an overshoot of 2.11%, reaching a maximum temperature of approximately 71.48°C, while the FO-PID_{MV} controller produces a nearly identical overshoot of 2.12%. The FO-PID_{GA} controller also reduces the settling time from 5.82 h for the classical PID and 7.42 h for the FO-PID_{MV} to 5.12 h, representing reductions of approximately 12.03% and 31%, respectively. Moreover, the ISE decreases from 6.81 for the classical PID controller and 5.74

Table 10 Performance indices of the classical PID and FO-PID controllers under the measured solar-current profile

Controller	Overshoot (%)	Settling time (h)	ISE (°C ² h)
PID	5.34	5.82	6.81
FO-PID _{MV}	2.12	7.42	5.74
FO-PID _{GA}	2.11	5.12	1.26

Table 11 Control-effort variation indices for the valve actuator under solar-current profile

Controller	$\sum_{k=1}^N \Delta u(k) $ (%)	$\max(\Delta u(k))$ (%)
PID	249.84	4.44
FO-PID _{MV}	1286.57	71.44
FO-PID _{GA}	1271.70	70.15

for the FO-PID_{MV} controller to 1.26 for the FO-PID_{GA} controller. These results indicate that the FO-PID_{GA} controller provides a better transient response and improved regulation performance relative to both benchmark controllers under the considered measured solar-current profile. Table 10 summarizes the corresponding performance indices.

The control-effort variation indices under the measured solar-current profile are presented in Table 11. The classical PID controller presents the lowest accumulated valve-command variation, with 249%, whereas the FO-PID_{MV} controller presents the highest value, with 1286%. The proposed FO-PID_{GA} controller presents an intermediate value of 1271%, which is higher than that of the classical PID but lower than that of FO-PID_{MV}. The maximum command increments are 4.44%, 71.44%, and 70.15% for the PID, FO-PID_{MV}, and FO-PID_{GA} controllers, respectively. Thus, the proposed controller requires greater valve activity than the classical PID but slightly less accumulated valve movement than FO-PID_{MV}.

4.5 Performance under Measurement Noise

The controller performance is assessed in the presence of measurement noise. The operating conditions and reference changes are the same as those considered in the setpoint-tracking test presented in Sect. 4.1. Zero-mean Gaussian noise with a standard deviation of $\sigma_n = 0.1^\circ\text{C}$ is added to the stack-temperature measurement. The same noise realization is used for all controllers to ensure a consistent comparison.

Figure 8 presents the resulting stack-temperature and valve-opening responses. The temperature responses remain similar to those obtained in the noise-free test, indicating that the measurement noise does not significantly affect the output-tracking performance of the controllers. Its effect

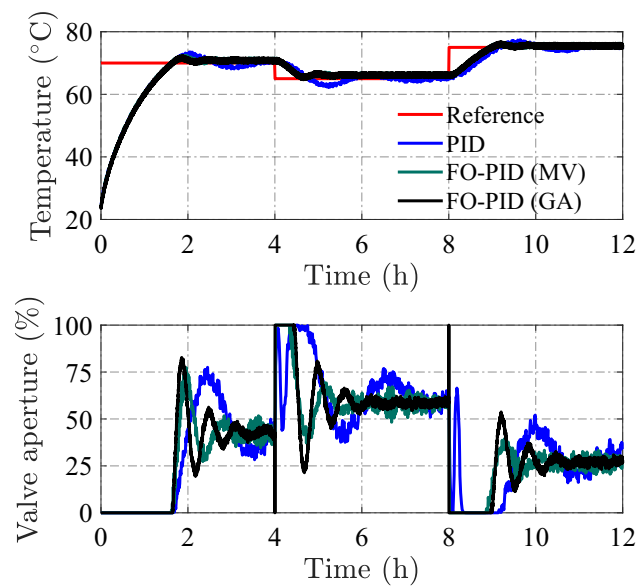


Fig. 8 Closed-loop responses under zero-mean Gaussian measurement noise with $\sigma_n = 0.1^\circ\text{C}$

Table 12 Control-effort variation indices for the valve-opening command under measurement noise

Controller	$\sum_{k=1}^N \Delta u(k) $ (%)	$\max(\Delta u(k))$ (%)
PID	2828.02	45.66
FO-PID _{MV}	4036.40	100
FO-PID _{GA}	7808.30	100

is more evident in the valve-opening commands, which exhibit additional sample-to-sample oscillations. During the constant-reference portions of the test, the maximum noise-induced peak-to-peak variation is approximately 5%.

Table 12 summarizes the accumulated valve-command variation and the global maximum command increment over the complete simulation. The accumulated variations are 2828.02%, 4036.40%, and 7808.30% for the PID, FO-PID_{MV}, and FO-PID_{GA} controllers, respectively. Therefore, although the output responses remain similar to the corresponding noise-free responses, the FO-PID_{GA} controller exhibits the greatest accumulated valve activity under measurement noise.

The global maximum increments reported in Table 12 include the larger valve movements associated with the reference changes and should not be interpreted as the amplitude of the noise-induced oscillations during steady operation. The latter remain approximately 5% peak-to-peak. These results complement the frequency-domain constraint imposed on the complementary sensitivity function. Although this constraint contributes to limiting measurement-noise transmission to the controlled output, it does not

directly limit noise amplification in the control command. Therefore, a practical implementation may require additional filtering, a rate limiter, or a small deadband to reduce unnecessary valve activity.

5 Conclusion

In this work, a fractional-order PID controller was proposed for the temperature regulation problem of an alkaline water electrolyzer, together with a genetic algorithm-based frequency-domain tuning methodology. The controller design was formulated from the transfer function obtained by linearizing the phenomenological plant model around a selected operating point, while the closed-loop assessment was carried out on the nonlinear model. The simulation results showed that the proposed FO-PID_{GA} strategy provides satisfactory tracking and disturbance-rejection performance over the considered operating conditions. The comparisons with a classical PID controller and an alternative FO-PID controller tuned using a frequency-domain methodology, adopted as benchmarks, indicated that the proposed FO-PID_{GA} controller provides improved overall closed-loop performance, as reflected by the settling time, integral of squared error, and overshoot results.

These results are particularly relevant for electrolyzer operation, since accurate temperature regulation is directly related to efficiency, safety, and component durability. In the long run, the ability of the proposed controller to keep the temperature close to the desired setpoint while attenuating transient deviations may contribute to reducing degradation effects associated with repeated temperature variations. Moreover, its disturbance-rejection capability suggests that it can support safe operation under the variations considered in the simulations, such as those induced by fluctuations in the power supplied by renewable energy sources. The measurement-noise assessment showed that the temperature responses remain close to their noise-free counterparts, although the accumulated valve activity increases, particularly for the FO-PID_{GA} controller. Therefore, additional filtering or a rate limiter may be required in a practical implementation.

Finally, it should be noted that the digital implementation of the FO-PID controllers relies on Muir's approximation, whose order directly affects the computational burden of the resulting discrete-time controller. Therefore, the approximation order should be selected so as to provide an appropriate compromise between approximation accuracy and implementation complexity.

Future work will consider experimental validation on a laboratory-scale alkaline water electrolyzer.

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Author Contributions Contributing authors: Styven Palomino (styven.palomino@posgrad.ufsc.br); Gustavo Artur de Andrade (gustavo.artur@ufsc.br); S.P. implemented the model, performed the controller tuning and simulation studies, and drafted the manuscript. G.A.A. defined the research methodology, supervised the work, contributed to the analysis of the results, and revised the manuscript. Both authors read and approved the final version of the manuscript.

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Data Availability No datasets were generated or analyzed during the current study.

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